MODEL PAPER I UNIT PHYSICS

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Year/Sem.:- I year / II Sem

Short Answer Questions

1. Write down important applications of Hall effect.
2. Explain, what is Meissner effect?
3. Explain, what is fermi level?
4. Distinguish between covalent and metallic bonds.
5. Write two properties of nano materials.

Section-B

Long Answer based Questions

1. Explain the formation of covalent Ionic and metallic bonding in solids.
2. Explain and deduce Bragg's law in x-ray diffraction.
3. The intrinsic carrier density at room temperature in Ge is 2.37 x $10^{19}/m^3$. if the electron and hole mobilities are 0.38 and 0.18 $m^2V^{-1}S^{-1}$ respectively. Calculate its resistivity.
4. The electron hole concentration in a sample of semiconductor are 5 $x10^{19}/m^3$ and $8x10^{20}/m^3$ respectively. If the mobility of electron and hole are $0.09m^2v^{-1}s^{-1}$ and $0.05m^2v^{-1}s^{-1}$ respectively, then calculate the hall coefficient of the semiconductor.
5. Describe nature and origin of various forces existing between atoms of solid crystals. Explain the formation of covalent, ionic and metallic bonding in solids.

Section-C

Long Answer based Questions
1.Explain energy band theory of crystals on its basis bring out difference between Conductors, Insulators and Semiconductors.
2. What is intrinsic semiconductor. Obtain an expression for the electrical conductivity of the intrinsic semiconductor.
3. (A) What is X-ray diffraction? Deduce Bragg's law for the diffraction of X-ray in a crystal, how Bragg's spectrometer is used to determine the wavelength of monochromatic X-rays?
(B) Assuming that there are $5x10^{28}$ atoms/m^3 in copper, find the Hall coefficient.
4. (a) The hall coefficient of certain silicon specimens found to be -7.5 x $10^{-5}m^3/C$ at a certain temperature. If the conductivity is found to be $200\Omega^{-1}m^{-1.}$ Calculate the density of charge carriers.
(b) What is Hall Effect ? Obtain the expression for Hall coefficient, Hall voltage and Hall mobility.